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Nadav Goldschmied , Philip Furley & Ruth Bush

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REVIEW



Critical review of uniform color effects in sports

Nadav Goldschmied^a, Philip Furley^b and Ruth Bush^c

^aDepartment of Psychological Sciences, University of San Diego, San Diego, CA, USA; ^bGerman Sport University Cologne, Institute Training and Computer Science in Sport, Köln, Germany; ^cHahn School of Nursing and Health Sciences, University of San Diego, San Diego, CA, USA

ABSTRACT

Groundbreaking research linking uniform colors and performance in sport has gained momentum in recent years demonstrating a superiority effect associated with certain colors, especially red, and an increase in aggression when in black. However, the findings are not uniform and some studies find no association. Our objective was to identify, compare, and disseminate a comprehensive list of studies associated with uniform colors and performance. This critical review identified 33 studies in sport, which explored the phenomenon and met the criteria for inclusion. We used the color-in-context theory (Elliot, A. J., & Maier, M. A. (2007). Color and psychological functioning. *Current Directions in Psychological Science*, 16(5), 250–254; Elliot, A. J., & Maier, M. A. (2014). Color psychology: Effects of perceiving color on psychological functioning in humans. *Annual Review of Psychology*, 65(1), 95–120) as the foundation for our analysis of the findings and, in turn, exposed some incoherence of some constructs and boundary conditions set by this framework. In the current work, we specifically distinguish between performance research and perception-only work and illuminate the differences as they pertain to theoretical underpinning. Based on present findings, we conclude that the evidence for uniform colors to influence the outcome of athletic competitions is weak and that more investigation is required to substantiate the effect, if it does exist. We conclude by proposing future avenues of research both as they pertain to features of the sports studied but also for the methodology utilized.

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Since Hill and Barton (2005) published their groundbreaking study on the superiority of red over blue uniforms in combat Olympic competitions and in quick succession Rowe, Harris and Roberts (2005) argued for a specific white disadvantage, a steady stream of research has emerged on the effects of uniform colors on sport performance. Regrettably, this growing body of research has yielded a highly ambiguous pattern of results. To bring clarity to the results, we have reviewed this research from a unifying theoretical perspective (i.e. color-in-context framework; Elliot & Maier, 2007; 2014) while highlighting methodological and theoretical shortcomings with the goal of advancing research practices and knowledge accumulation within this field.

Color effects on human perceptions and functioning is not a new domain of research. Color is omnipresent and long held public beliefs (Goldstein, 1942; Hardin, Hardin, & Maffi, 1997; Mogensen & English, 1926) as well as research link it to psychological functioning. For example, numerous studies associate the color red to attraction enhancement (Elliot & Niesta, 2008; Guéguen & Jacob, 2013; Schwarz & Singer, 2013) yet a recent comprehensive meta-analysis disputed the effect (Lehmann, Elliott, & Calin-Jageman, 2018).

The perception of color is the product of its wavelength (Fairchild, 2013). Other color dimensions are lightness (i.e. white-to-black property) and chroma, which is reflected in the vividness of the color experience. Uniform color studies in sport focus exclusively on the first element but ignore the other two important elements. This is not a minor omission because some sports are played in open-air stadiums (e.g. soccer, football) in which visibility might be obstructed by rain or fog and thus blunt color effects.

Uniform colors play a ubiquitous role in almost all sports distinguishing competitors from each other. The need to fundamentally differentiate visually between opponents is of utmost importance for players to separate 'friend from foe' in team sports, for officials to make the right calls and for spectators both in the arena as well as on TV (or other digital means) to enjoy the game and their fandom. Without this fundamental contrast, sports is almost impossible to play. As such, both amateur and professional leagues have strict policies and rules about who wears what (Benoit, 2019), including hefty fines for violations.

So strong is the association between certain sport entities and uniform colors (usually the home kit) that it has given rise to popular speculations in regards to performance. For example, Manchester United was accused of trying to influence the outcome of the game by switching their uniform colors when they were behind in half time (Wright, 2017). Superstitious beliefs in the National Football League connect The Dallas Cowboys' blue jersey uniforms with subpar performance most likely due to their defeat at Super Bowl V, the crowning game, in 1971 (Lukas, 2007). Alternatively, the legendary golf player, Tiger Woods, always competes in red on Sundays during the final round of tournament play. He is believed to be performing especially well then under stressful circumstances (Kershaw, 2019).

Another motivation behind the attempt to review the scientific literature in the field of uniform colors and performance comes from the extensive coverage it received from the popular and influential press outlets such as the BBC, CBS and the New York Times (Dambeck, 2008; Jones, 2020; Leach 2009; Schwartz, 2005; Stafford, 2012). This journalistic reporting supposedly based on overwhelming academic research may propel the lay-public to assume that the phenomenon is empirically well established and extensive.

An earlier review of color effects included a short sports research subsection (Elliot & Maier, 2014) but did not expand on studies in the field. More critically, it omitted theoretical implications as they pertain to this unique context and did not elaborate on research deficiencies across the domain, which may call into question some of the conclusions reached.

Theoretical foundation of uniform color research in sport

Elliot and Maier (2007, 2014) lamented the absence of a theoretical foundation to ground color effects in earlier research. In fact, the only theory offered before the color-in-context

(CIC) comprehensive theory (Elliot & Maier, 2012) was that of Goldstein (1942) who presumably associated wavelength with arousal, whereupon longer-wavelength colors (e.g. red, yellow) were experienced as stimulating and brought about outward focus while shorter-wavelength colors (e.g. blue, green) promoted inward orientation as well as reserved and more consistent behavior. However, since Goldstein's ideas were developed strictly based on his work with abnormal clinical patients, was demonstrated in simple perceptual processes rather than interactive behavioral/ performance effects, and was unclear in terms of the specific hypotheses it generated, CIC served as the prism through which we attempted to interpret the relevant research.

The CIC theory (Elliot & Maier, 2012) has six core assumptions: (1) Colors go beyond aesthetics and convey specific information. (2) The meaning of the color information is grounded in our evolved biology as well as learned associations. Some colors carry a unique meaning, which is acquired through biological orientation reflected in evolutionary processes. Subsequently learned association through cultural conditioning may take place to reinforce the effect (for example, linking red with caution through 'stop' signs for drivers or red error markings by teachers). Elliot and Maier argued that evolutionary principles had a stronger impact over cultural associations. (3) Perceiving color triggers evaluative processes automatically (i.e. rudimentary processing to discern 'good vs. bad' or 'foe vs. friend') at an early stage of perceptual processing. (4) The triggered evaluative processes serve to motivate rudimentary forms of behavior (e.g. approach or avoid). (5) The influence of color on psychological functioning is assumed to be automatic (i.e. without conscious intent or deliberate processing). However, Elliot and Maier posit that there is a reciprocal relationship—meaning that affect, cognition, and behavior can have an influence on color perception. (6) The meanings of colors and their effects on psychological functioning and behavior are contextual, for example, red may evoke an approach motivation in a courtship setting but an avoidance orientation in achievement/ sport context.

We focus our critical review (Grant & Booth, 2009) on studies in the field of sports, which we define in accordance with the International Olympic Committee as competitions based on physical athleticism or physical dexterity. In line with the inclusion criteria of the Olympic Committee (Furley, 2019), we exclude noncompetitive physical activities (i.e. exercise) and strategic games that mainly depend on chance (e.g. board games) as well as games based on decision making without significant physical undertaking (e.g. chess). This inclusion criteria led us to exclude studies that pertain to the broader achievement settings such as in the academic or strictly cognitive domains (IQ testing in Elliot, Maier, Moller, Friedman & Meinhardt, 2007; Stroop task in Ioan et al., 2007; anagrams in Mehta & Zhu, 2009) as well as games (poker in Ten Velden, Baas, Shalvi, Preenen & De Dreu, 2012; arcade/ computer games in Recours & Briki, 2015; Sorokowski & Szmajke, 2011). We also excluded research that studied environmental colors and not uniform ones (Etnier & Hardy, 1997; Keller & Vautin, 1998) as in that context color is not necessarily identified with the opposing side and may convey a different interpretation.

Method

An extensive literature search was conducted using PsycINFO and the Web of Science for the search terms 'uniform colors', 'sport' combined with 'athletic performance', 'winning', 'red', 'blue', 'black' and 'aggression,' to identify empirical articles published, in English,

before July 2020. Relevant articles' reference lists were hand searched to identify any applicable scholarly works such as dissertations, theses, unpublished work and abstracts not listed in the searchable databases for inclusion in the review. Inclusion criteria were broad; studies needed to include methodology to detect a difference in performance or perceived performance based on uniform color as the hypothesis and publish quantitative results. This process continued until the uniform color research pool was exhausted.

Once the studies were identified, the authors used critical review methodology to summarize and to evaluate the strengths and weaknesses of each study's methodology and results as well as to present an evaluation of the total body of literature on uniform color (Grant & Booth, 2009). Two independent reviewers screened all identified articles that contained quantitative assessment of performance, regardless of expertise or gender (see our PRISMA flow chart to monitor our progression). Thirty-three studies met the criteria. The colors studied, sport, sample size (when available), sample characteristics, and effects were compiled for evaluation and comparison and presented systematically in Table 1 (Figure 1).

This critical review approach is inclusive and permits a single critical framework to address many different perspectives and theories within one review. Additionally, the review methodology permitted critical evaluation of innovative thematic classifications identified during the review process: Performance and Game/Winning Outcome Research; Color Perception Research in Sports and Color Effects on Aggression in Sports. We consider this methodology the most suitable review approach for integrating and summarizing existing empirical research on color effects in sports with CIC, gaining an understanding of an ambiguous body of research, and laying the groundwork for theory development and methodological advancement in this field (see Grant & Booth, 2009). In addition, this methodological approach allowed for conceptual innovation and inclusion of multiple sports for comparison, bringing further diversity and sample size to a typically narrow analysis. The resulting outcome was a synthesis of existing approaches of looking at the continuum of uniform color effects throughout the entirety of the domain.

Uniform color research in sport

In sports, uniform color effects mostly focused on red over other colors (Meier, Hill, Elliot & Barton, 2015). It is argued, in line with CIC theory, that the color red is advantageous for the athlete wearing red and that this advantage is grounded in both biological pre-dispositions and/or past conditioning experiences. The former is rooted in evolutionary history of intra-sexual rivalry linking red coloration to dominance and a tendency to aggress. The link has been demonstrated in correlational research with nonhuman primates (Bergman, Ho & Beehner, 2009), observational exploration in nature (Khan, Levine, Dobson & Kralik, 2011), and experimental work. For example, utilizing random assignment painted male red Gouldian finch birds won about 80% of their dyadic contests against un-colored juvenile competitors. In contrast, black and blue counterparts managed only the expected base-rate of 50% (Pryke, 2009). The effect may be driven by real changes in aggression when in red (Healey, Uller & Olsson, 2007), opponents' intimidation and subsequent avoidance (Setchell, Charpentier & Wickings, 2005) or both (Pryke, 2009). Research of human physical performance in a non-sport setting might be taken as further evidence for the biological predisposition reasoning by demonstrating a muted response when observing red (termed as a 'threat cue') before performing a maximal voluntary

Table 1. Authors, colors studied, sport, sample size and characteristics and effect found for the studied included based on the criteria definition.

Authors	Colors Studied	Sport	Research design / Sample Size	Sample Characteristics	Effect
<i>Performance and Game/Winning Outcome Research</i> Hill and Barton (2005)	Blue, red	Olympic combat sports	Archival/ Men (n not specified); women (n=155)	2004 Olympic games	Red advantageous over blue in Olympic competitions when competitors of similar strength. Color association not evident in power asymmetries.
Rowe, Harris, and Roberts (2005)	Blue, white	Olympic Judo	Archival/ 301 combats/	Natural experiment stratified by strength asymmetries categories	Blue beat white at Olympic level at medium asymmetry level; no association at other levels.
Matsumoto, Konno, Hata, and Takeuchi (2007)	Blue, white	Judo	Archival/ 1623 bouts	World Championships 2001,2003,and 2005 and 2004 Olympiad	Blue associated with wins among men but not among women.
Attrill, Gresty, Hill, and Barton (2008)	Red, non-red	Soccer	Archival/ 56 seasons (n games not specified)	Elite soccer league; 1946–2003 seasons	Red uniforms playing at home more successful. No effect at away venues and in non-red home uniform. Contrasting teams playing at same city, red consistently outperformed non-red uniforms in their average league position.
Dijkstra and Preenen (2008)	Blue, white	Judo	Archival/ 501 matches	Controlled for allocation biases, asymmetries in prior experience and differences in recovery time. Hypothesis: blue associated with being stronger competitor	No color effect on winning.
Kocher and Sutter (2008)	Blue, red, darker	Soccer	Archival/ 306 games	Assess uniform color effects separately for home and away	Color had no association with winning either home or away
García-Rubio, Picazo-Tadeo and González-Gómez (2011)	Red, non-red	Soccer	Archival/ Spanish Football League season 2001/02 - 2008/09	Examine budgets and team resources versus game outcomes	Red had no effect on outcome when budget and attendance were accounted for.
Piatti, Savage, and Torgler (2012)	Red, non-red	Rugby	Archival/ 5604 games	30 seasons of elite Australian rugby	Red associated with success in descriptive analysis, but not associated after multivariate analysis.
Dreiskaemper, Strauss, Hagemann, and Büsch (2013)	Various	Artificial combat	Experimental/ 28 males	Quantify heart rate and fighter strength	No differences in fight performance as function of color. Participants wearing red had higher heart rates during fight (not prior or after the fight) and produced more force before match.

(Continued)

Table 1. Continued.

Authors	Colors Studied	Sport	Research design / Sample Size	Sample Characteristics	Effect
Pollet and Peperkoorn (2013)	Black, white, red, blue, and 'other'	Ultimate Fighting Championship	Archival/ 210 matches/ archival	Video review of non-Olympic combat sport	No color effect on winning, including analysis by referred and undisputed outcome.
Greenlees, Eynon, and Thelwell (2013)	Red, green, blue, yellow	Soccer	Experimental/ 40 males	Experienced soccer players took penalty kicks against goal keepers with randomly assigned colored jerseys.	Players scored fewer goals against red versus blue or green. No difference compared to yellow.
Allen and Jones (2014)	Red, non-red	Soccer	Archival/ 7720 games	1992–93 to 2011–12 UK soccer teams	Red associated with improved average league position. No effect between red and home winning percentage.
Noel Furlay, Hüttermann, Nopp, Vogelbein, and Memmert (2014)	Red, non-red	Soccer	Archival/ 322 penalty kicks	Examined effects of color of goal keeper and of penalty kicker during penalty shootouts	Neither kicker nor goalkeepers performed better wearing red compared to different colored jerseys.
Julio, Miarka, Rosa, Lima, Takit and Franchini (2015)	Blue, white	Judo	Archival/ 1233 competitions	Sex, age group, and level of competition were also considered in the analysis	Blue judogi may be associated with positive competitive outcome for both sexes in regional and state level competitions and for athletes above junior age
Falcó, Conchado, and Estevan (2016)	Blue, red	Taekwondo	Archival/ 65 combats	Female, elite competitions; colored electronic head and body protectors	Red significantly associated with winning in featherweight category; did not generalize to other categories
Vasconcelos and Del Vecchio (2017)	Red, black	Wushu Sanda	Archival/ 46 fights; 55 athletes	Women randomly assigned to wear red or black	Red won 65% of fights. Statistically significant but small sample.
Dijkstra, Preenen, and van Essen (2018)	Blue, white	Judo	Archival/ 45,874 contents	Used a 2011 rule change, which reversed the color of the top-seeded judogi from blue to white for analysis	Found as color of top-seeded judogi changed, winning uniform by color changed similarly.
Goldschmied and Lucena (2018)	Blue, red, white	Basketball	Archival/ 186 games	Six seasons of national tournament games; excluded matches with yellow or other light colored uniforms other than white	No color effect on winning.
Seife (n.d.)	Blue, red	Olympic combat sports	Archival/ Not given	2008 Beijing Olympics secondary data	Red not advantageous
Goldschmied and Spitznagel (2020)	Blue, red, black, white	Basketball	Archival/ 407 games	Seven seasons of women national tournament games; excluded matches with yellow or other light colored uniforms other than white	No color effect on winning.

<i>Color Perception Research in Sports</i> Greenlees, Leyland, Thelwell, and Filby (2008)	Soccer	Experimental/ 12 experienced male, semi-professional soccer goalkeepers	Video presentation to measure impressions formed by opposing goalkeepers of kickers wearing red or white.	Penalty takers wearing red were perceived to possess more positive characteristics than those wearing white but color seemed to be of a peripheral influence . Red competitor awarded average of 13% more points than blue competitor.
Hagemann, Strauss, and Leifing (2008)	Taekwondo	Experimental/ 42 experienced referees	Referees manipulated the colors red and blue in a counter-balanced experimental design.	Wearing red enhanced perceptions of relative dominance and threat.
Feltman and Elliot (2011)	Taekwondo	Experimental/81 (2 studies)	College student perceptions of clothing watching video on computer monitor	No difference in impressions of players dressed in red vs. white or a neutral point-light condition (no color).
Furley, Dicks, and Mummert (2012)	Soccer	Experimental/ 22 Male soccer goalkeepers (study 1)	Assessing pre-performance behavior before a penalty kick of experienced German players.	No difference when comparing blue, green, red, and yellow uniforms. When limited to red vs. blue, referees and participants with rules expertise judged tackles from behind more harshly for players wearing red.
Krenn (2014)	Soccer	Experimental/ 42 experienced male referees	Examine uniform color and judging of the severity of tackle using video	In both countries, black and red trunk wearing boxers were perceived erroneously to land more blows against a blue competitor, but not enough to influence the outcome of the fight.
Sorokowski, Szmajke, Hamamura, Jiang, and Sorokowska (2014)	Boxing	Experimental/ 290 students from Poland and Mainland China	Cross-cultural analysis of Polish and Chinese participants; watch two-minute video of a semi-professional boxing bout with manipulated boxers' trunks	Competitors wearing red in boxing and wrestling were perceived to be more aggressive and more likely to win than their rivals wearing blue or green uniforms. No differences were detected in taekwondo.
Krenn (2015)	Boxing, wrestling, taekwondo	Experimental/ 210 sports students	Digitally manipulated uniform color of contestants.	No difference related to uniform color
Krenn, Pernhaup, and Handstreiner (2017)	Soccer, handball	Experimental/ Soccer (39); handball (118)	Within-subjects design to measure psychological and behavioral dimensions.	Teams wearing black more likely to receive penalties and spend time in the penalty box than non-black uniform teams.
<i>Color Effects on Aggression in Sports</i> Frank and Giovich (1988)	Hockey	Archival + experimental / 40 football fans + 20 referees (study 3) and 72 male students (study 4)	Study of wins and penalties during 1970– 1986 National Football League (NFL) & National Hockey League (NHL) seasons	No significant differences between black uniforms and others.
Tinyaki (2005)	Soccer	Archival / 2142 games	Study of yellow and red cards and penalty kicks in seven seasons Turkish Premier Soccer League.	
	Softball			

(Continued)

Table 1. Continued.

Authors	Colors Studied	Sport	Research design / Sample Size	Sample Characteristics	Effect
LeMaire, Short, Ross-Stewart, and Short (2007)	Black, blue, red, yellow, green		Experimental/ 120 female collegiate players	Between-subjects design to measure psychological perceptions	Team shown in red was rated significantly higher in confidence compared with those in blue and yellow and overall 'strongest.' The black uniformed team was rated the 'meanest.'
Caldwell and Burger (2011)	Black/non-black; red/non-red	Hockey	Archival / 102 games black vs. no black, 61 red vs. no red	2008–2010 NHL seasons allowed use of third color jersey and chance to contrast same team interactions with different colors.	Neither black nor red uniforms were related to penalties or home winning percentage.
Webster, Urland, and Correll (2012)	Black; non-black	Hockey	Archival / 52,098 games	25 seasons of NHL penalty-minute data	Teams wearing black were assessed more penalty minutes. White jerseys were uniquely associated with less aggression in contrast with colored uniforms.

contraction of the thigh in contrast to a prior exposure to chromatic or achromatic control color (Payen, Elliot, Coombes, Chalabaev, Brisswalter & Cury, 2011).

Given the purported evolutionary importance of the color red in agonistic encounters, it is not surprising that the majority of research in the context of sports has studied the influence of the color red. Hence, we first focus on studies on the color red in sport contexts using CIC as theoretical background. We utilized the following taxonomy to classify studies that fell within the selection criteria: (1) performance studies exploring winning the competition, almost exclusively archival in nature (2) experimental research studying the effects of color perceptions on both affect and cognition, and (3) studies exploring aggressive behavior (archival and experimental) specifically with black.

Performance and game/Winning outcome research

Studies in support of red uniform effects. Studies detecting red uniform effects span several sports, which vary in duration, aggression-proneness, composition (individual vs. team), and target (ball vs. opponent).

As stated before, the pioneering study to demonstrate this association was conducted by Hill and Barton (2005). They showed that when contestants in Olympic combat sports (i.e. boxing, Taekwondo, Greco-Roman, and free wrestling) were assigned to wear red (and not blue) in a purported natural experiment setting during the Athens 2004 Olympic Games, they won significantly more than the expected 50% of competitions. Furthermore, when the researchers took into account differences in strength between the competitors based on quartiles of points difference, the effect was evident only when the contestants were equal to each other but not in the small to large power asymmetries. They concluded that the red color effect did not overwhelm the competition but provided a slight yet robust enough advantage to tilt the outcome when balanced initially. It is also of note that Barton and Hill (2005) failed to detect the effect in Olympic women's competitions with a limited sample size ($N = 155$).

In women's taekwondo bouts of four elite competitions including the London Olympics in 2012, Falcó, Conchado, and Estevan (2016) studied the effect of colored electronic head and body protectors (rather than full sport uniforms). They found a significant association between donning red (as opposed to blue) and winning the competition in the featherweight category (though the sample size included only 65 combats), which failed to generalize to other weight categories.

Vasconcelos and Del Vecchio (2017) studied the outcomes of Wushu Sanda, a mixed orientation combat sport during the 13th female World Championships in which competitors duel while being randomly assigned to either wear red or black uniforms. They found that those wearing red won 65 percent of their fights. However, the results from this study need to be interpreted with care due to the small sample size.

Attrill, Gresty, Hill, and Barton (2008) extended the research of Hill and Barton (2005) to the field of soccer—a long duration team, ball-oriented sport in which aggression plays a much lesser role on game outcome. The researchers showed in an exhaustive set of archival studies that English soccer teams wearing red uniforms playing at their home venue during the 1946–2003 seasons were significantly more successful than other teams as well as more likely to win championships altogether. Commensurately, they did not find any

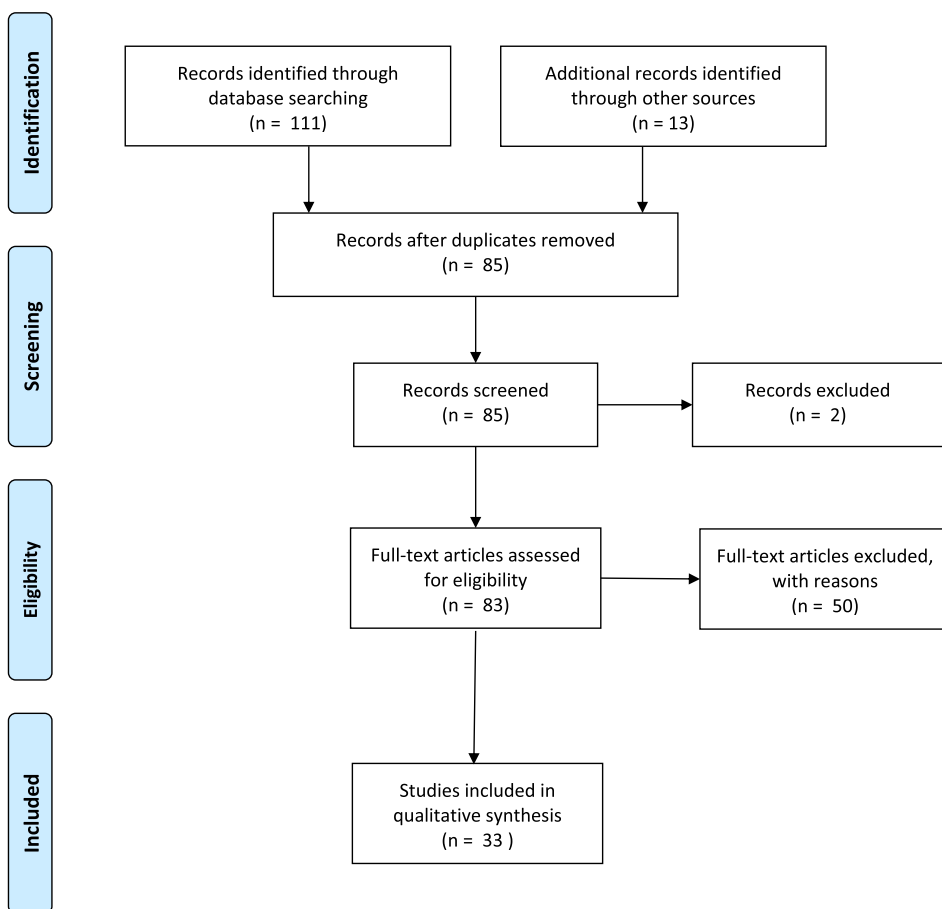


Figure 1. Prisma flow diagram.

differences in performance when the very same teams were playing at away venues when their uniform was different than their red ‘traditional’ home uniform.

Also using the English Soccer Premier League, Allen and Jones (2014) relied on a more recent game sample (1992/93–2011/12, $N = 7720$) some of which corresponded with Attrill et al. (2008). They found that while red wearing teams had an improved average league positioning than non-red competitors, they failed to detect the effect in home winning percentage. The researchers struggled with how to reconcile the contrasting findings and reasoned that either there was no effect altogether or that

‘... changes in psychological functioning associated with viewing or wearing red stay with team members after shirt color has been changed. In English professional football, teams are known for wearing particular colors and it is possible that team members become conditioned to associate certain colors with certain teams – the team then becomes the catalyst for greater levels of threat’ (p. 13).

Piatti, Savage, and Torgler (2012) explored the red effect in Australian rugby, a team ball-oriented game with a substantial aggression element during a period extending 30 seasons ($N = 5604$ games) for the 36 teams competing in the country’s elite league.

They found the effect in a direct binomial contrast. This finding seems especially impressive since the teams in the sample did not use one color but a majority/ minority hue scheme, which should have diminished the effect. However, when utilizing advanced multivariate statistics, the effect was no longer apparent suggesting that it may have been driven by a spurious association (i.e. very successful teams wearing red by happenstance). The researchers went on to argue that 'This could explain Attrill et al.'s (2008) results in the English Premier League as Liverpool, Manchester United, and Arsenal are historically some of the most successful teams and are all wearing red.' (p. 16).

Of note, the research undertaken in soccer and rugby, aside from being team sports, is unique in being conducted outdoors which may influence visibility and color perception in lights of sunshine, rain or clouds.

Lastly and in departure with previous archival correlational research, Greenlees, Eynon, and Thelwell (2013) conducted a controlled experiment. Forty experienced soccer players took 10 penalty kicks against a goalkeeper wearing a black jersey and a week later, took additional 10 penalty kicks against a goalkeeper randomly assigned to wear either a red, green, blue, or yellow jersey. The players were asked before each set of kicks about their expectancy of success. Players facing a red uniformed goalkeeper scored fewer goals ($M = 5.4$) than those facing either blue ($M = 7.2$) or green-clad ($M = 7.5$) goalkeepers (with no difference to yellow uniforms), yet their expectations to score were similar across conditions. This might be interpreted in line with the CIC theory (Elliot & Maier, 2012), which assumes that potential color effects on behavior operate largely outside of conscious deliberation.

While the number of studies in support of the red-superiority effect in sport performance is limited, several distinct sports were investigated. Another strength is the longitudinal method used to demonstrate the effect (Attrill et al., 2008; Piatti et al., 2012). However, this line of research also shows weaknesses as most of the performance studies find only qualified support for uniform color effects depending on narrow guidelines and very specific circumstances (Allen & Jones, 2014; Falcó et al., 2016) or when using one statistical approach but not another (Piatti et al., 2012). In addition, many studies fail to detect the effect altogether. We review these studies in the next section.

Studies failing to detect effects of red on performance. In combat sports, Seife (n.d.) attempted a replication of Hill and Barton (2005) groundbreaking study in the subsequent Olympics and failed to support the effect. He also fundamentally disputed the claim made by the authors that their study was a true natural-experiment, an assumption at the heart of their analysis and interpretation. Although a limitation of this work was that it never underwent peer-review, the methods and conclusions of the manuscript appear well-grounded.

Pollet and Peperkoorn (2013) investigated the effects of wearing different colored trunks (the only article of clothing) in Ultimate Fighting Championship (UFC), a non-Olympic combat sport with very minimal rules and thus possibly the most similar to fighting in nature according to the researchers. In 210 international bouts, they failed to find an effect of wearing the color red on winning percentage of fighters. An additional analysis contrasting competitions decided by referees and ones which ended undisputed (i.e. Technical Knockout, Knockout, submission or tap – out) still failed to find any color effects.

Dreiskaemper, Strauss, Hagemann, and Büsch (2013) conducted an experiment in which an artificial combat situation was created while randomly assigning different color sport attire to the participants. No differences were found in performance assessed by raters as a function of clothing color. However, the researchers measured the heart rate

and strengths of the fighters prior to and during the bout and found that participants wearing red had higher heart rates during the fight (not prior or after the fight) and were able to produce more force prior to the confrontation.

Moving beyond individual combat sports, Kocher and Sutter (2008) in German soccer studied performance during the 2000/1 season of the top division. Wearing red or blue at home (or away) did not catapult teams to winning more in comparison to the average league winning percentage in the home and away locales. In the Spanish Professional Soccer League (2001/02–2008/09 seasons), García-Rubio, Picazo-Tadeo and González-Gómez (2011) also found that when endowments of resources (i.e. game attendance) and clubs' budgets were accounted for, teams in red did not win more frequently than other teams. Lastly in soccer, Noel, Furley, Hüttermann, Nopp, Vogelbein, and Memmert (2014) analyzed all penalties from penalty shootouts during European and World Championships from 1982 to 2012. They found no red advantage for goalkeepers wearing a red jersey ($N = 22$), nor for penalty takers wearing red ($N = 72$).

In basketball, another ball dominated, long-duration team sport, Goldschmied and Lucena (2018) explored the performance of American collegiate teams during the national annual tournament in a period spanning six seasons (2012–2017). In this sporting event, higher ranked teams usually wear white uniforms while lower seeded competitors may choose the color of their uniforms. The researchers failed to detect any color effect in a multi-color comparison. In a follow-up study, Goldschmied and Spitznagel (2020) could not detect an effect in women's collegiate basketball as well.

Taken together, it seems that for each study that supported the red superiority effect in sport, there is one to contrast it. This confusing state of affairs could be addressed by considering both theoretical and methodological issues arising from the studies. Before we turn to these concerns, we first review the remaining studies that examined different colors other than red, but fall within our selection criteria.

Studies in support of white uniform disadvantage effect. Another line of research points to a distinct white uniform disadvantage (Julio, Miarka, Rosa, Lima, Takito & Franchini, 2015; Rowe et al., 2005). This research, though, is not grounded in CIC theory (Elliot & Maier, 2012) but is based on a visibility argument. These studies assume that white stands in unique contrast to the background and thus movements initiated by the white clad athletes can be detected earlier on and be better anticipated by the opponent. No special meaning is attributed to white and the sole basis for the disadvantage is based on the distinct contrast properties of the color.

In contrast with the red superiority effect, white disadvantage was strictly examined in the sport of judo due to the now established tradition of using blue and white as the standard uniform colors (Matsumoto, Konno, Hata & Takeuchi, 2007).

Rowe et al. (2005) studied judo competitions and found that blue clad competitors beat their white wearing counterparts significantly more than the expected 50% chance in what they argued amounted to a natural experiment. However, when broken-down into strength asymmetries, the effect was manifested only at the medium level but not in the absent, small, and large difference categories. The researchers explicated that a white judogi had an enhanced contrast against the background and thus the wearer's moves were foreseen earlier and consequently better defend against. Hill and Barton, in a rebuttal, contended that the visibility explanation was improbable as contestants fought in very close physical proximity (mostly holding each other) in brightly lit venues.

Matsumoto et al. (2007) citing the limited sample size and methodological problems of Rowe et al. (2005), explored the effect in three World Championships (2001, 2003 and 2005) as well as the original data set of the Athens 2004 Olympics. Similar to the original study, they found a prevailing and growing blue advantage effect (53.54% in the first round to 60.22% in the final medals stage) but no effect for the women's competitions.

Julio et al. (2015) were the first ones to study uniform color effects in the youth ranks. They conducted an extensive analysis of competitions in regional and state youth judo tournaments. They found that when no seeding system was utilized, white was at a disadvantage, including in girl's competitions. However, the bias was manifested only in the older age groups (17 years old and older) but not with younger competitors. The researchers proposed that uniform color provided only a slight advantage and since differences in maturation were quite large at a young age, the white disadvantage could have been manifested only in an older age when skill variability was reduced.

Studies failing to detect white uniform disadvantage effects. In an unswerving effort, Dijkstra and Preenen (2008) challenged the assumptions made by Rowe et al. (2005) that their findings were based on a natural experiment. They demonstrated that stronger seeded competitors were more likely to wear blue in the early rounds of the tournament (Rowe et al. later attempted to remedy their original analysis by eliminating the first round of the 2004 judo competition). Persisting in their criticism, Dijkstra and Preenen found that the bias still extended beyond the initial round. They argued further that this possible confound may have also influenced resting time between bouts. When they excluded all bouts which involved a seeded athlete ($N = 104$), they failed to detect a white disadvantage effect. They subsequently examined 501 matches, which constituted the gold-medal finals of 72 elite tournaments with equal resting times. Blue judogi competitors (winning percentage = 50.7%) performed similarly to white ones. Lastly, Dijkstra and Preenen underscored their contention that wearing blue was associated with being a stronger competitor by showing that following a rule change, which increased the number of seeded judogi in elite competitions, the association between wearing blue and winning was strengthened.

Dijkstra, Preenen, and van Essen (2018) continued their dogged archival pursuit of debunking the white disadvantage utilizing a 2011 rule change, which reversed the color of the top-seeded judogi from blue to white. In an extensive data set, they found that with the reversal, the winning uniform by color shifted similarly.

Color perception research in sports

The second line of research on colors in sports investigated the mechanisms of how color perception could potentially influence the behavior of athletes. Perception studies are usually conducted as an experiment in the laboratory and differ on the type of presentation offered such as still photos or videos as well as the composition of the participants involved, ranging from student convenience samples to referees (novice or experienced). They also vary on the task the participants were asked to undertake ranging from imagining themselves as competitors to volunteering opinions as to what their rivals project (e.g. dominance, status). Yet other perception research requested participants to make judgments of performance acting in the role of game referees (e.g. was an infraction committed?).

Red superiority in perception. Most basically, Krenn (2015) digitally manipulated uniform color (blue, green or red) in photos of two contestants competing in three combat sports. Competitors wearing red in boxing and wrestling were perceived to be more aggressive and more likely to win than their rivals wearing blue or green uniforms. No differences were detected in taekwondo possibly due to the minimal chromatic body surface in this sport.

Utilizing a self-involved strategy, Feltman and Elliot (2011) examined the influence of red vs. blue attire in taekwondo on perceptions of dominance and threat prior to a supposed about-to-happen same-sex competition. Two experiments assessed the effect for oneself and for the opponent among undergraduate students who were shown the clothing on a computer monitor. The effect was bidirectional in nature as wearing red enhanced perceptions of relative dominance and threat while observing an opponent in red enhanced perceptions of the opponent's relative dominance and threat (the participants seemed to be unaware of the influence). These effects were observed across gender. LeMaire, Short, Ross-Stewart and Short (2007) also digitally mastered opponents' photos and found similarly the perception of red superiority in projected confidence (over royal blue and yellow) with an all-female sample of NCAA softball players.

Greenlees, Leyland, Thelwell, and Filby (2008) used a video presentation to examine the impact of soccer penalty kickers' uniform color and gaze focus before taking a penalty kick on the impressions formed by opposing goalkeepers. Soccer goalkeepers observed footage filmed from between the goalposts of a player preparing to take a penalty kick. The kicker wore either red or white. Penalty takers wearing red were perceived to possess more positive characteristics than those wearing white (e.g. confidence, assertiveness).

Sorokowski, Szmajke, Hamamura, Jiang, and Sorokowska (2014) were the first to conduct a cross-cultural analysis and had participants from both Poland and China watch a two-minute video of a semi-professional boxing bout. The researchers manipulated the colors of the boxers' trunks (red, black, and blue) and the participants viewed one of all six possible versions of the same fight. In both countries, black and red trunk wearing boxers were perceived erroneously to land more blows against a blue competitor though not sufficiently to influence the outcome of the bout. The researchers were the first to argue that black provided an advantage in sport (aside from merely being grounded in aggression proneness) and explained their findings in evolutionary processes as the meaning attributed to red in Chinese color was not in line with its Western connotations. Also in combat sports, Hagemann, Strauss, and Leißing (2008) used a similar manipulation in tae-kwon-do with experienced referees manipulating the colors red and blue in a counter-balanced experimental design. Regardless of identity, the competitor wearing red was awarded an average of 13% more points than the competitor wearing blue.

Lack of support for red superiority in perception. Krenn, Pernhaupt, and Handsteiner (2017) analyzed the role of red and blue uniforms in two experiments utilizing a within-subjects design. Participants were asked to rate penalty takers in soccer and penalty takers and goalkeepers in handball on various psychological (e.g. confidence) and behavioral (e.g. where the ball would be placed) dimensions. No consistent evidence indicating a beneficial influence of red uniform colors was evident.

Furley, Dicks, and Memmert (2012) attempted a replication of Greenlees et al. (2008) assessing the pre-performance behavior of those about to take a soccer penalty kick. The results, in contrast to the original study, showed no difference in the impressions formed between players dressed in red vs. white or a neutral point-light condition (no color) by experienced German goalkeepers. The authors suggested that the color associations in England (the original study) and theirs may be different based on the success of the National team: Germany traditionally in white and historically very successful in penalty shootouts while England in red and historically very unsuccessful in penalty shootouts (Jordet, 2009). Yet the replication did not exhibit a white advantage as the original red superiority effect but merely no differences in perception.

Also in soccer, Krenn (2014) studied whether uniform color (achromatic and chromatic) had an influence on judging the severity of tackles by both referees and non-referees utilizing digitally manipulated video clips. No differences arose when comparing blue, green, red, and yellow uniforms. More limited contrast of red vs. blue only, showed that referees and participants with a high level of rules knowledge judged tackles from behind more harshly for players wearing red.

While red superiority in perception has been demonstrated in several sports (combat and non-combat), research failing to establish the effect were almost exclusively conducted in soccer.

Color effects on aggression in sports

The third line of sport research contends that uniform colors can consistently influence aggressive behavior of athletes or at least the perception thereof. More specifically, it is assumed that wearing black uniforms is associated with higher levels of on-field aggression (Frank & Gilovich, 1988; Webster, Urland & Correll, 2012). Frank and Gilovich first proposed that through cultural pairing in the US, black has acquired an association with malevolence and thus professional players are both more aggressive as well as perceived as such by others (without making any predictions in regards to superior athletic performance).

Studies in support of black uniform increasing aggression effect. Congruent with CIC theory (Elliot & Maier, 2012) of color meaning and subsequent motivational shifts, Frank and Gilovich (1988) demonstrated that teams wearing black in the National Football League (NFL) and the National Hockey League (NHL) were perceived as more malevolent (LeMaire et al., 2007 for a similar effect). They also found that black clad teams were more likely to lose ground due to penalization or spend more time in the penalty box, respectively. In the same vein, they showed that in two professional teams switching from a non-black to black uniform showed immediate increase in penalties incurred.

In another study, these researchers also presented experimental evidence through a choreographed football play in which they manipulated the uniform colors of the defensive team. When in black, observers (both students and referees) assessed a higher likelihood of committing an aggressive infraction than in non-black outfit for the same performance, in support of perception as a contributing factor to color effects. Lastly, Frank and Gilovich brought participants together and randomly assigned them to either wear black or not. When in black, teams selected games to supposedly compete against another team that were more aggressive in nature.

Webster, Urland, and Correll (2012) expanded on the earlier study and explored an impressive 25 seasons of NHL penalty-minute data (from all 30 teams). They replicated the original findings as teams wearing black were assessed more penalty minutes. Also, for the 10 teams which switched to- or from black uniforms, the effect was marginally significant (possibly due to reduced power). Additionally, the researchers also found that white jerseys were uniquely associated with less aggression in contrast with colored uniforms.

Studies failing to detect the effect of black uniform on aggression. Titling their study 'On thin Ice', Caldwell and Burger (2011) also studied hockey in the NHL but focused on a unique set of circumstances. During two seasons (2008-2010), the league allowed teams for a limited number of home games to wear a third jersey aside from the customary dark at home and light away. Eight teams opted to wear black while six others choose red. This shift allowed to contrast their performance against the same opponent with different uniform colors. While studying a limited set of games, the authors found no support that either black or red uniforms were related to higher levels of aggression (e.g. minutes penalized, severe penalties) or superior performance (e.g. home winning percentage).

Tiryaki (2005) scrutinized the aggressive behavior (e.g. yellow and red cards distributed as well as number of penalty kicks) in the Turkish Premier Soccer League for the soccer teams wearing black vs. other uniform colors during seven seasons (1995-2003). No significant differences emerged between teams wearing black uniforms and others. However, it remains unclear how this lack of an effect might be explained given potential cultural differences between Turkish athletes and athletes from Western countries. It might be the case that black is not similarly associated with malevolence as in Western countries.

How might the ambiguous findings on color effects in sports be explained?

For every study finding an effect of color in sports, there seems to be a study disputing it. This begs the question of what might explain the ambiguous pattern of results. One simple explanation might be that the impact of colors on athletes might not be strong enough to reliably affect performance in the context of sports. However, this explanation is in contrast with the substantial number of studies that claim to have found empirical support for color effects on sports performance. Although, it seems unreasonable to assume that so many published studies could simply be wrong, there currently is a reason to believe that problematic research and publication practices have led to low rates of replications and weaker empirical support for psychological theories than researchers want to believe (Lakens & Evers, 2014; Munafo et al., 2017; Nelson et al., 2018). Some authors even go as far as to announce a 'crisis of confidence' (Pashler & Wagenmakers, 2012, p. 528). Regarding color research in sports, two methodological shortcomings that have been discussed abundantly as major factors leading to published effects that are not true seem particularly problematic in this field: appropriate sample sizes (Button et al., 2013; Fraley & Vazire, 2014; Schönbrodt & Perugini, 2013; Schweizer & Furley, 2016) and the related issue of researcher degrees of freedom (e.g. Simmons, Nelson, & Simonsohn, 2011), which focuses on the dilution of findings based on the researchers' methodological decisions. Numerous examples compellingly demonstrate

this leads to inflated rates of false-positives or Type 1 errors (Simmons et al., 2011). A false-positive means that researchers report finding an effect that does not actually exist. In other words, researchers reject the null hypothesis when it is true (Simmons et al., 2011, p. 1359): 'despite empirical psychologists' nominal endorsement of a low rate of false-positive findings ($\leq .05$), flexibility in data collection, analysis, and reporting dramatically increases actual false-positive rates'. Many of the reviewed studies on the effects of color in sports had limited power to detect true effects and therefore have a higher chance of reporting false-positives. Regarding limited degrees of freedom, the reviewed literature in favor of color effects often found these effects after using very particular ways of data analysis.

Utilizing this critical statistical prism, it is clear that some domains of uniform color effects hold better than others. For example, the white visibility disadvantage effect seems to be not robust enough as the research debunking it (Dijkstra & Preenen, 2008; Dijkstra, Preenen & van Essen, 2018) which is much more comprehensive in scope than the one making the original claim (Rowe et al., 2005). On the other hand, the association of black and aggression seems more promising, with studies in support (Frank & Giolovich, 1988; Webster, et al., 2012) being longitudinal and having substantially larger samples than those not showing the effect (Caldwell & Burger, 2011; Tiryaki, 2005).

Beside these methodological issues, there might also be theoretical particularities of CIC (Elliot & Maier, 2012) that explain some of the ambiguity of findings on uniform color in sports. A basic tenet of CIC states that contextual variables need to be taken into account when investigating color effects in real world scenarios. For example, CIC assumes that associative learning can enhance or diminish effects of certain colors. From this follows that the color black might trigger aggression in participants from a certain cultural background, but not in others (compare e.g. the results from Frank & Giolovich [1988] and Tiryaki [2005]). In addition, CIC assumes that the meaning of colors can also be grounded in evolved biology. Therefore, it has been suggested that the meanings of certain colors can potentially vary as a function of sex. In this respect, some of the diverging findings, for example on the color red on performance in combat sports between men and women (Hill & Barton, 2005) could be explained by biological differences in processing colors. But context may be even more specific pertaining to one sport, as Chantal and Bernache-Assollant (2017) found that yellow formed negative impressions of racing cyclists (but not in other racing sports) because the researchers argued, with years, this color assumed supposedly the meaning of opportunism and cheat from frequent pairings with doping (possibly termed otherwise as the Lance Armstrong effect).

Another important assumption of CIC is that the effects of colors are often assumed to be automatic (occurring at an early stage of processing), but can potentially be inhibited by intentional processing (premise 5 of CIC). Therefore, the lack of a hypothesized color effect in a certain sport could be explained by athletes using particular cognitive strategies to override an automatic effect of perceiving a certain color on a behavioral or psychological outcome. In this respect, a study (Bertrams, Baumeister, Englert, & Furley, 2015) showed that available self-control capacity moderated the debilitating effect of red on performance. More specifically the study showed, that the color red only had an effect in a performance situation (intelligence testing) when participants had limited self-control capacity, whereas red did not affect performance when self-control capacity was intact. Transferred to the sport context, this might suggest that fatigued athletes

(little self-control capacity) might be more prone to color effects, while athletes that are rested are less prone. Taken together, the above examples demonstrate CIC allows researchers plenty of flexibility in interpreting potential color effects or the lack thereof in the context of sports. While, CIC can be considered a useful theoretical framework for structuring and integrating a vast amount of color research, it can be critiqued as a too flexible theory in deriving precise hypotheses and predictions since it can account for almost any pattern of data. For example, the lack of a color effect in a specific test might be attributed to particular constraints of the context (e.g. combat vs. team sport) or participants (e.g. sex or cultural background) or the state of athletes (fatigued vs. rested).

Discussion

Elliot and Maier (2014) lamented that their CIC theory was missing real-world generalizable research to support it empirically. As such archival studies, which explore performance in highly competitive sport settings, should be of great value in validating (or negating) the comprehensive framework they proposed. Elliot and Maier (2014) concluded: 'The data appear to be mixed, but the most recent, and perhaps methodologically strongest, research yielded supportive evidence' (p. 102). Unfortunately, based on the current critical sport review, we do not share their position and the totality of the evidence is far from impressive. By our count among published peer-reviewed research and without correcting for the problem of publication bias (Franco, Malhotra, & Simonovits, 2014), about half of the available studies debunk the effect (see Table 1) while highlighting serious methodological problems which were outlined in the previous sub-section. Furthermore, studies that detect the effect (e.g. red superiority: Hill & Barton, 2005–505 citations on google scholar, October. 4th 2020), tend to be quoted extensively while those that do not (Pollet & Peperkoorn, 2013–9 citations on google scholar, October. 4th 2020), gain fewer attributions [journal and year of publication notwithstanding].

The original theorists (Elliot & Maier, 2014; Maier, Elliot, Lee, Lichtenfeld, Barchfeld & Pekrun, 2013) propose that this domain of research is in its nascent stage. We concur, but the current state of affairs leads us to doubt the possible sway uniform colors have over sport performance (and subsequently game outcome). Potentially, uniform colors influence perceptions as studied in laboratory research but one has to acknowledge that impression formation requires a significantly lower level of 'force' overall to yield significant results while game outcome is the product of reduced/elevated athletic output (in addition to the initial impression formation). In this regard, possibly, the refereeing element is most plausible in uniform color influence since it requires only perception effects to impact the score rather than physical/performance changes, yet this path has not been studied sufficiently with expert officials to be evaluated comprehensively.

In light of the accumulated research we outlined in this review, several major clarifications are required in regards to the CIC theory if it to account for the totality of the data. First, it should clarify how the evolutionary based associations (Pryke & Griffith, 2006) work in conjunction with cultural association mechanisms (Frank & Gilovich, 1988). While the two surely reinforce each other, can the effect be sustained separately? Or even when they operate antagonistically? For example, Goldschmied and Spitznagel (2020) who surveyed the research field and showed that in the laboratory with female majority samples,

red produced performance effects time and again (Elliot et al., 2007; Elliot & Aarts, 2011; Feltman & Elliot, 2011; LeMaire et al., 2007) but in their archival study of NCAA women sports, a null effect emerged. This contrast is of importance since evolutionary explanations of intra-sexual rivalry should not play a role for females though cultural based associations should work equally well for both sexes.

The articles reviewed have focused on color study context, color perception, and color impact on competitive sports with less emphasis on the methodology to establish and the statistical analysis of the relationship between color and winning. The literature examining uniform color and winning is heterogeneous in the extreme with little uniformity in participants, methodology (including study duration), or statistical approach. Participants were diverse and include team members as well as individuals; expertise ranged from competing as juniors at regional meets to college students watching video to elite Olympian to professional athletes. Often the exact ages or specific skill levels were not included in the published methodology or were quite varied. For example, English Premier League players in 1947 were not necessarily comparable to players in 2003, but were evaluated as if they were (Attrill, et al., 2008). The participants were primarily, but not exclusively male. Also, studies differ on type of color comparisons they conducted. The original Hill and Barton study (2005) showed red superiority specifically over blue, Attrill et al. (2008) showed the red effect without regard to the opponents' uniforms and Goldschmied and Lucena (2018) studied the same effect contrasting red and blue against white.

Much of the data on which the studies were based was retrospective and observational, predominantly reliant on sport-result archives with few interventional studies. Uniform color was assigned by seeding, history, and randomly depending upon the sport, rule changes (which are not always fully known to the researchers), and in interventions by the investigator. Generally, the outcome or dependent variable was 'win' but winning was not measured uniformly. For example, 'win' included mean rank in home league, mean percent wins obtained at home, and mean number of points per game at home (Attrill et al., 2008). To complicate matters even more, some game outcomes include also ties and not only a winner and a loser (Attrill et al.). The majority of the studies report descriptive statistics and rely heavily on non-parametric ranking statistics with only few using a multivariate approach to control for potential correlates such as home versus visitor field, weather conditions, or even team payroll. Without reported means, correlations, or Cohen's D, there were few chances to compare effect (size) among the studies.

Finally, Field and Gillett (2010) suggest that a random-effects approach should be part of social science data analysis. Much of the sport and color literature does not provide the needed context to do so, which is partially because research publication format. Several of the most notable studies were not published as research manuscripts having undergone a complete peer-review process but rather as a brief communication (Hill & Barton, 2005) or as a letter to the editor (Rowe et al., 2005; Pollet & Peperboom, 2013).

Although pointing out problems in a particular area of research is an important first step for progress, the more important second step is creating viable practices and guidelines for future research. In this respect, we consider the open science guidelines emerging from the 'crisis of confidence debate in psychological science' a good way of strengthening the quality of empirical sports color research in the future (Lakens & Evers, 2014; Munafo et al., 2017; Nelson et al., 2018). More specifically, future research

in this area needs to pay attention to appropriate sample sizes (Button et al., 2013; Fraley & Vazire, 2014; Schönbrodt & Perugini, 2013; Schweizer & Furley, 2016), the avoidance of researcher degrees of freedom (e.g. Simmons, Nelson, & Simonsohn, 2011), preregister plans for studies and data analyses (e.g. Nosek, Ebersole, DeHaven, & Mellor, 2018), be transparent in a prior hypotheses and data analyses (e.g. Nosek et al., 2015) that have been precisely derived from a solid theoretical foundation, conduct replications (Camerer et al., 2018; OSC, 2015), make sure that an effect does not only apply to a highly specific data set but to the phenomenon of interest (Fiedler, 2011), and, perhaps most importantly, a heightened awareness for the theoretical and methodological challenges in color research in sports.

More concretely, we recommend that future sports color research should take the following concrete steps to reduce the problems identified. The study should be planned as detailed as possible by, for example, using an established pre-registration template (e.g. van't Veer & Giner-Sorolla, 2016). When filling in the template, researchers should pay particular attention to (a) precisely formulate a priori hypothesis based on sound theory or designate the study as exploratory; (b) Pre-specify the context and sample and provide a solid rationale why the chosen context and sample is suitable for testing the hypothesis; (c) calculate a priori sample size (Faul, Erdfelder, Lang, & Buchner, 2007); (d) publish the findings no matter what the outcome of the study is; and (e) update the theory based on the study results.

Before outlining recommendations for future research, we want to acknowledge that our literature review is not without limitations as we only included published reports in English and did not reach out to obtain unpublished results on color effect (e.g. Master thesis or doctoral dissertations). Further, the chosen critical review approach might be more subjective as other approaches and we cannot exclude the possibility that other researchers would draw different conclusions from the available studies as was for example demonstrated in the project 'many analysts, one dataset' (MAOD; Silberzahn et al., 2018).

Future Directions in uniform color effects research

Besides implementing steps to improve methodological practices, we further suggest that researchers should sharpen their research questions. Using sports performance or even competition outcome as dependent variable (as many sports color studies have attempted) seems problematic. Sports performance is influenced by a wide range of interacting factors and is highly difficult to predict. Therefore, we consider it fruitful to take a step back and systematically investigate into the effects of color on processes arguably related to sports performance: e.g. on maximal force production, endurance, motor coordination, risky decision making, or play creativity.

Further, in the age of big data (Morgulev, Azar & Lidor, 2018), archival research should be continuously updated to assess if earlier findings stand the test of time. Future researchers should take the mantle repeatedly in regards to Olympic combat sports performance, fouling rates in the NFL and NHL as well as NCAA basketball neutral site games winners.

Another avenue of research could take advantage of unique times which bring about change to tradition. For example, the NFL from 2009 to 2017 had an October month-long

breast cancer raising awareness campaign in which players wore accessories in pink (Vrentas, 2016). Past rudimentary research (Profusek & Rainey, 1987) linked environmental pink with calm and thus the potential for reduced aggression in sports. Alternatively, the 2020 pandemic caused NBA teams to play in one location (thus eliminating travel and home-court familiarity) without any fans in the stands (only virtual ones) while allowing the designated 'home team' to wear its traditional home-kit. Tracking the home field winning percentage under these extraordinary times should be informative. All in all, researchers should always be on the alert for rule changes or extenuating circumstances to study uniform color effects.

Lastly, an important methodology to quantify research areas are meta-analyses. Therefore, we consider the priority of future research on colors in sports to improve research practices as outlined before conducting a meta-analysis. Meta-analysis requires an analysis question drawing upon studies having similar participants, intervention, comparators, and outcomes (Schardt, Adams, Owens, Keitz, & Fontelo, 2007) so that the data are reasonably pooled and the resulting summary estimate is meaningful (Basu, 2017).

In conclusion, through this work, we advocate cautiousness when interpreting color effects in sports and hope to improve future research in this fascinating field of study.

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